

Final Projection Report

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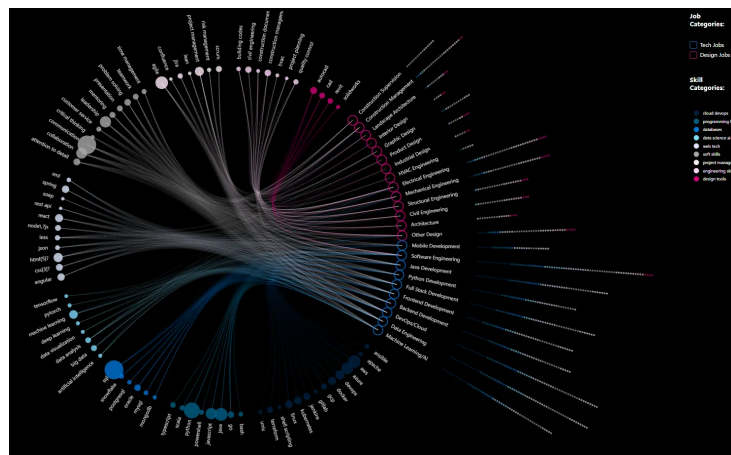
Context: the dilemma for cross-field job seekers

The labor market is evolving at an unprecedented pace. Emerging technologies and shifting demands are driving individuals to acquire new skills and transition into unfamiliar fields. This phenomenon is particularly evident among cross-field job seekers—those venturing into careers that differ significantly from their previous experience. While these individuals bring diverse perspectives and creativity, they often face challenges in aligning their skills with job requirements and navigating the vast job landscape.

In this study, we aim to provide a data-driven understanding of the broader categories of jobs and how specific skills, particularly in tech and design, connect to these roles. We analyzed LinkedIn job postings from 2023, extracting skills from job descriptions and categorizing them into nine overarching skill groups (e.g., "Data Analysis," "Programming") and 23 job categories (e.g., "Cloud DevOps," "Software Engineer," "Product Management"). Using a consistent keyword mapping process, we ensured clarity and accuracy in linking skills to jobs. Our visualizations focus on revealing the connection between tech/design skills and job categories, as well as providing insights into salary trends and geographic distribution across these roles.

Key Results

Observation 1: Preference for Both Design and Technical Skills Varies Across Job Categories



Circular Skill-Job Linkage Interactive Graph

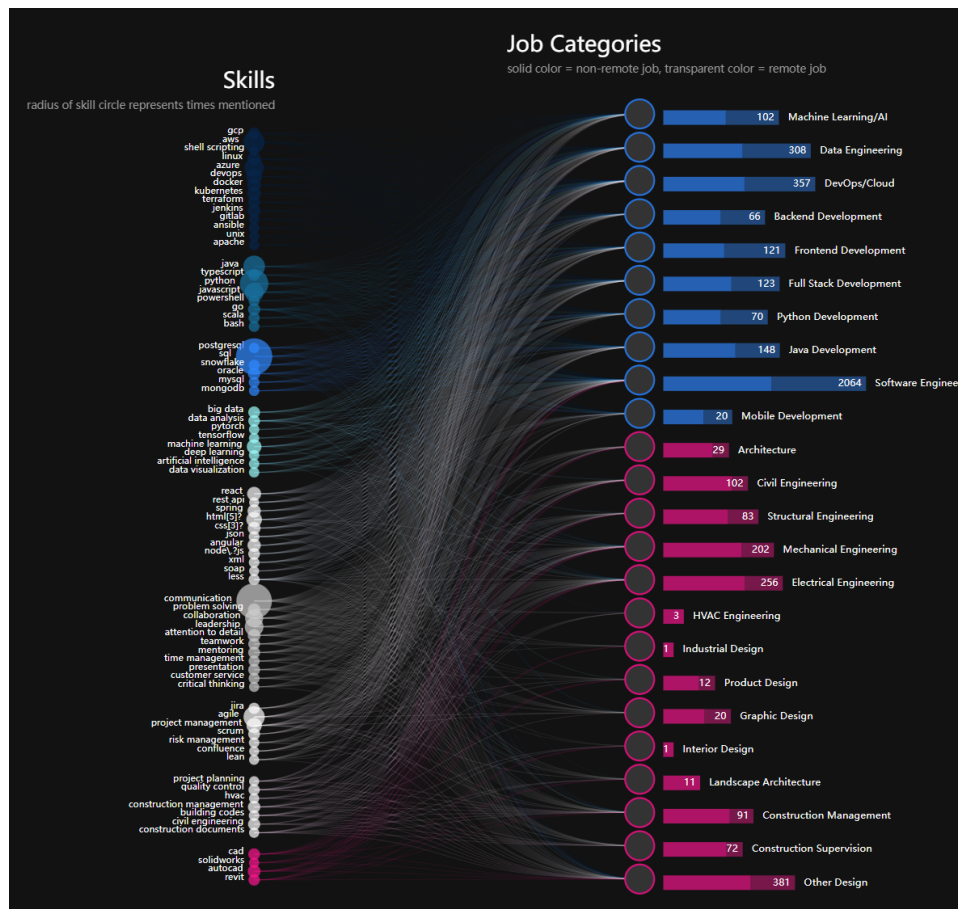
- From the visualization, we observe that certain roles, such as **Software Engineer**, demonstrate a strong preference for both technical and design skills, evident from the dense network of connections linking both types of skills to these jobs. In contrast, other roles, like **Interior Design**, predominantly emphasize design-specific skills, with little to no overlap with technical skills.
- Overall, technical jobs tend to require a broader and more diverse set of skills compared to design jobs. This discrepancy could stem from:
 - **Tech Stack Variability:** In technical fields, the differentiation among tools and frameworks is more pronounced. For instance, there are over 20 variations in software scaffolding tools alone (e.g., React,

Angular, Node.js), whereas the design field focuses primarily on a few key software solutions like Adobe and Figma.

- **Data Bias:** The dataset itself contains fewer design job postings than technical roles, leading to the omission of less frequently mentioned design skills (filtered out for having fewer than 50 mentions).

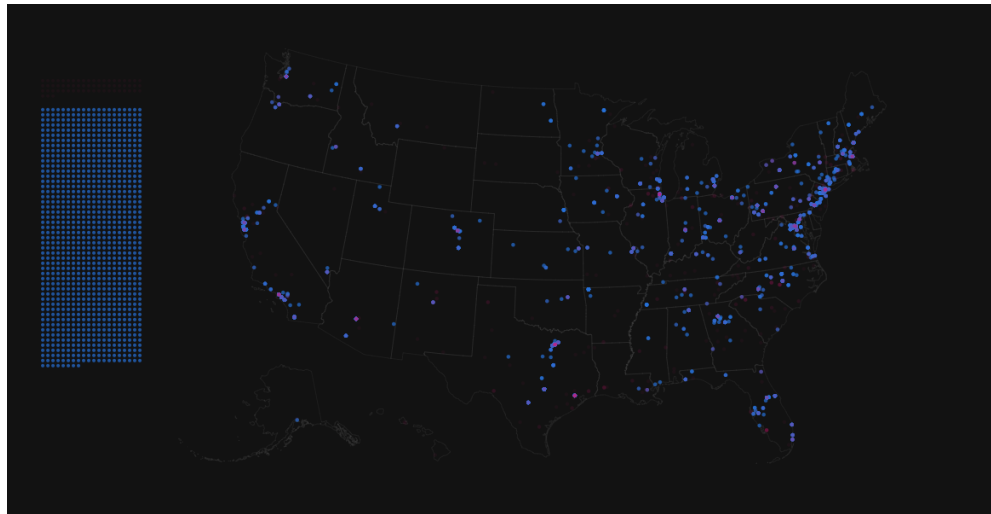
Observation 2: Overall tech-related jobs have more job openings comparing to engineering or design related job types, and provide more over half of the tech-related jobs are remote, while about less than 25% of the design related jobs are remote.

- To compare tech and design-related jobs by data count, we mapped the total job counts in subcategories and their remote positions. We used a log scale to prevent software engineering jobs from skewing the results. The data shows that tech jobs significantly outnumber design job openings, with about half of tech positions offering remote work compared to less than 20% for design jobs. Some design job categories have no remote positions in our sample dataset. Software engineering leads the tech category with over 2,000 job postings, while electrical and mechanical engineering are the largest design categories. While our dataset isn't comprehensive and requires further refinement and validation, some categories show very low posting numbers—just 3 for HVAC engineering and 1 for interior design.

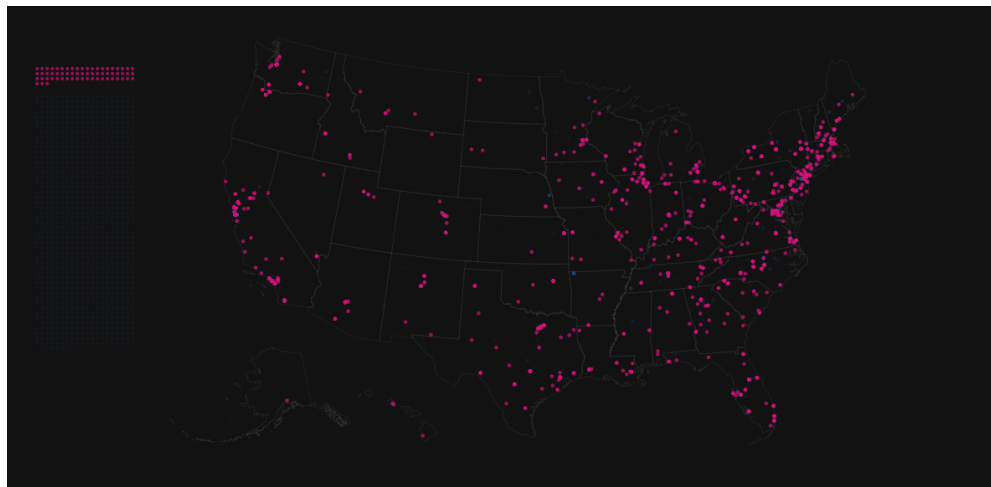


Batch-selectable Skill-to-job-categories connections graph

- Geographically, both design and tech jobs are concentrated in major cities along the east and west coasts, with the highest concentration on the east coast. Design jobs appear more frequently in the Midwest, where tech jobs are scarce. Design positions can also be found in smaller cities that lack tech job opportunities.



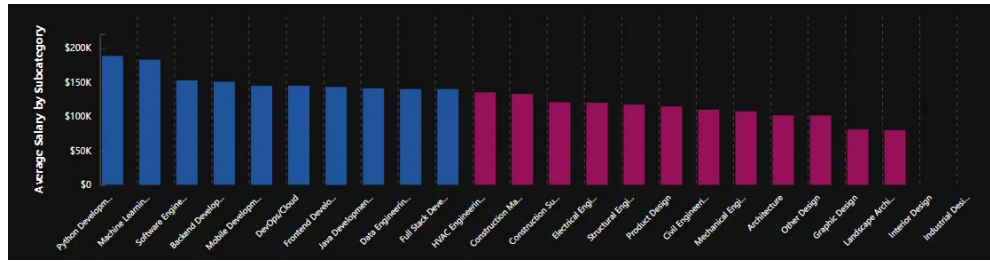
remote and non-remote jobs geolocations for tech-related postings



remote and non-remote jobs geolocations for design-related postings

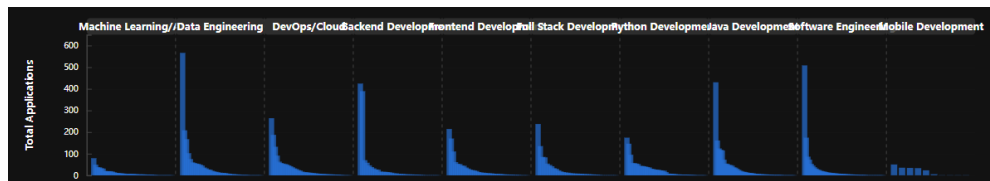
Observation 3: Overall, tech jobs has higher overall salaries and demonstrate more competition than design related jobs.

- According to the average salary by subcategory graph, tech-related job categories command the highest average salaries, followed by design-related categories. Backend-related and algorithmically complex jobs offer the highest average salaries, with compensation decreasing toward front-end positions. Within design jobs, technical engineering roles pay the most, while less technical positions like architectural designer and graphic designer have the lowest average salaries.

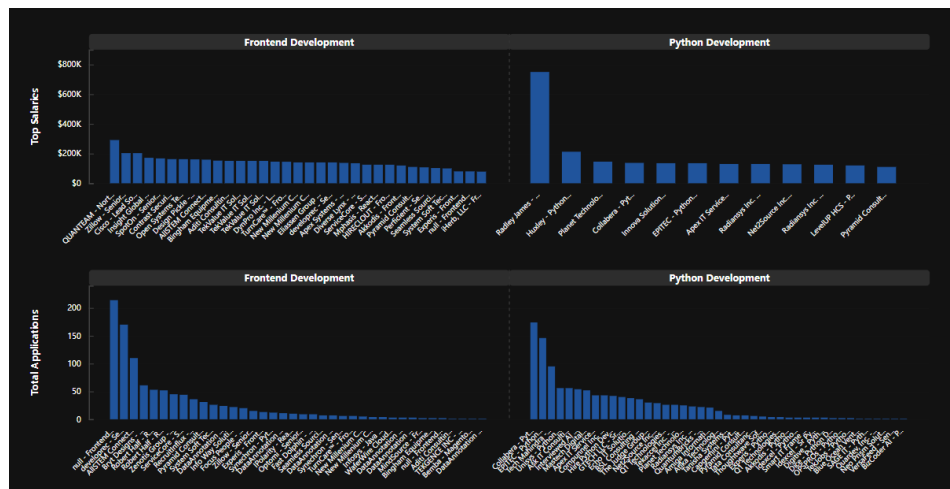


Average Salary bar chart by job subcategories

- Regarding application volume per position, some tech job categories receive up to 500 applicants for a single role. Many tech subcategories, including frontend development and Python development, have at least 25% of their positions attracting more than 50 applicants each.

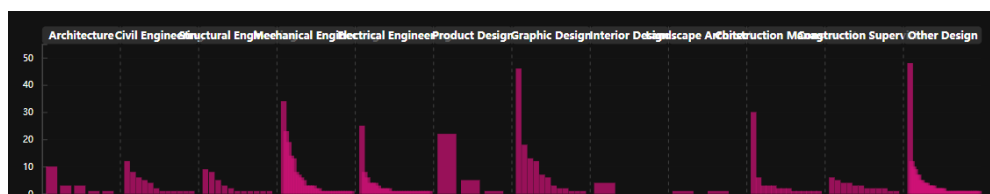


Filtered Tech Jobs Applications for job postings



filtered job category salary and total applications for two tech-related job categories

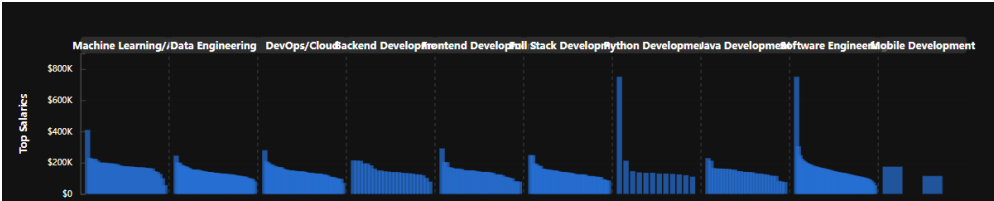
- For design jobs, the highest number of applications per posting is 50, with the first quartile showing around 10 applications per opening. While our limited dataset may affect these numbers, the general trend suggests design positions attract fewer applicants than tech roles.



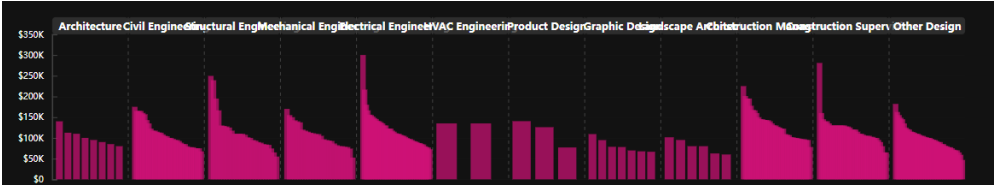
Design-related job's application quantity by job categories

- The salary distribution analysis shows stark differences between tech and design jobs. Top-paying tech positions can exceed \$600,000 annually, with most roles clustering around \$150,000. In contrast, design jobs cap at

approximately \$300,000, with the majority falling below \$100,000.



Tech-related job's salary distribution by job categories



Design-related job's salary distribution by job categories

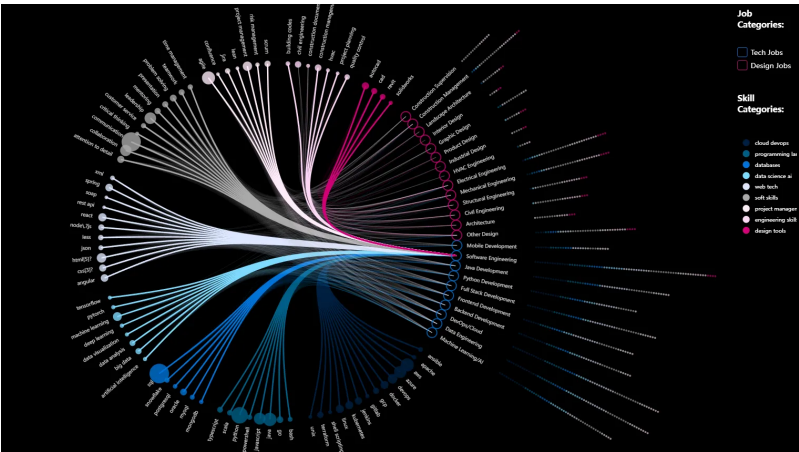
Design Decisions:

Color Coding for Clarity and Accessibility

Distinct colors were used to differentiate between skill categories (e.g., tech, design tools) and job types (tech jobs in blue, design jobs in pink). This ensures that users can quickly identify connections and trends at a glance. Additionally, the consistent color scheme across the dashboard makes navigating complex data intuitive, reducing cognitive load.

Interactive Link Generation

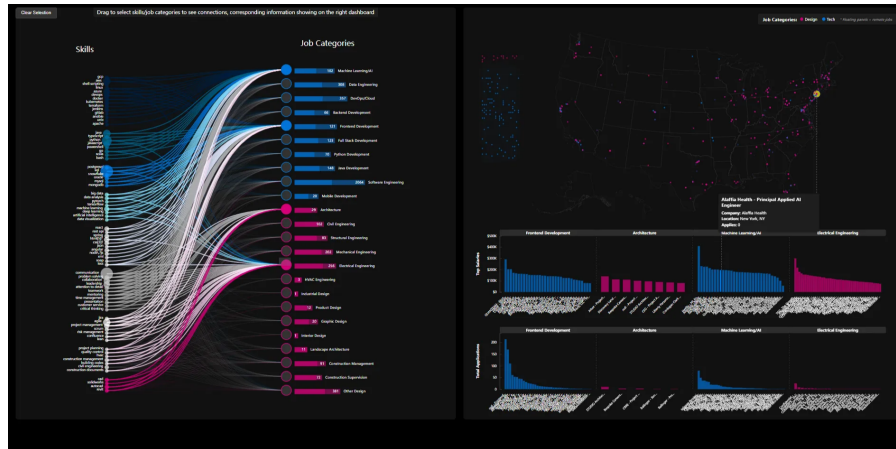
The hover and multi-select functionalities enable users to focus on specific nodes and highlight their direct connections. This allows users to dynamically explore the relationships between skills, job categories, and other variables. On the main panels, we intentionally displayed only high-level categories to avoid information overflow. However, users seeking detailed insights can hover over nodes to view specific job postings, clearly presenting more granular information while keeping the overall visualization uncluttered. This balance ensures both macro-level clarity and micro-level depth, making the visualization adaptable to different user needs.



Circular Skill-Job Linkage Interactive Graph

Integrated Dashboard Linking Skills, Jobs, Geo-Info, and Salary

In the final dashboard, multiple data dimensions are seamlessly connected, integrating skills and job categories with geographic distribution and average salary information. This allows users to explore relationships holistically—such as the connection between specific skills, job types, and their geographical and salary distributions. The inclusion of interactive linking enables users to see how salaries and job locations are interconnected, making it easier to grasp the overall trends and variability in the job market. Users can compare jobs across categories and use the skill-job connections to understand how different locations impact compensation and opportunities.



Interactive main dashboard

Reflection:

This project combined personal experience and technical collaboration to address challenges faced by cross-field job seekers. The idea was inspired by Risa's journey as a design background student transitioning into the tech field, with support from David, who shared similar experiences. Together, we sourced and cleaned the dataset, with David leading data preprocessing. Visualization tasks were split: Risa created the edge bundling diagram, David handled the GeoMap, and both collaborated on the skill dashboard. Integrating the dashboard with the GeoMap proved challenging due to differing data structures, so we delegated the task entirely to David while Risa focused on earlier parts of the page.

The project deviated from our original plan to analyze long-term trends due to the lack of comprehensive historical data, leading us to focus on high-quality data from the most recent year. This shift provided a clear snapshot of the current job landscape. Looking back, better upfront planning for consistent data sources and reusable components would have streamlined the process. Moving forward, we aim to apply these lessons in future projects, emphasizing a unified global design and deepening our understanding of tools like Svelte and JavaScript for more efficient collaboration and storytelling.